

Polynomial regression

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [3]: X=6*np.random.rand(100,1)-3
y=0.5*X**2+1.5*X+2+np.random.randn(100,1)
```

```
In [4]: X
```

```
Out[4]: array([[ -1.20725939],
               [ -1.78591175],
               [  1.80868505],
               [-0.39861747],
               [  2.61424862],
               [  0.84393264],
               [  2.18882298],
               [  0.23945412],
               [  1.61656944],
               [  2.48264388],
               [  1.41373659],
               [  0.32027881],
               [  1.10336752],
               [-1.2204481 ],
               [-1.65524983],
               [  1.92232145],
               [  1.05214745],
               [  0.5396229 ],
               [  0.23440715],
               [  0.08465351],
               [-1.79102278],
               [-0.71963614],
               [-0.52242988],
               [  2.14505865],
               [-1.6672044 ],
               [  2.40423067],
               [  1.25065832],
               [-2.07368344],
               [-1.69015877],
               [-1.94923723],
               [-0.12118408],
               [-1.66560648],
               [  0.25063486],
               [-1.09697935],
               [-2.82097769],
               [  0.03549212],
               [-1.37204019],
               [  0.96304763],
               [  0.04759388],
               [-2.74125073],
               [-1.67652677],
               [  0.88107259],
               [  2.06400014],
               [  2.38706241],
               [-2.25577764],
               [-2.51137703],
               [  2.7553812 ],
               [-1.75043904],
               [  2.46133893],
               [  0.76228407],
               [-2.01108896],
               [-0.16249032],
               [-2.62795553],
               [  1.72636103],
               [-2.16728004],
               [  1.45243762],
               [-0.86227157],
               [-1.4163866 ],
               [  0.44948477],
               [-0.34696528],
```

```
[-1.4194468 ],  
[-1.98598454],  
[ 1.71822107],  
[ 1.94095467],  
[ 1.20334984],  
[-0.84853658],  
[-0.66814752],  
[-2.67173265],  
[-1.72163424],  
[-2.69625501],  
[-0.04858207],  
[-0.7273757 ],  
[-2.78106339],  
[ 1.36806687],  
[-0.13090741],  
[-0.42653444],  
[ 1.90655367],  
[-1.90471326],  
[-0.4492429 ],  
[-1.09685704],  
[ 1.06102604],  
[-2.1497267 ],  
[-2.46081753],  
[ 0.35708945],  
[ 2.50583967],  
[ 0.06773081],  
[ 0.06685965],  
[-1.68890996],  
[ 0.18158089],  
[-2.6035114 ],  
[ 2.77923627],  
[-2.2764764 ],  
[ 1.55237529],  
[ 2.39982527],  
[-2.47949416],  
[-2.39573445],  
[-2.62158334],  
[ 0.14169308],  
[ 2.59171718],  
[-0.35941172]])
```

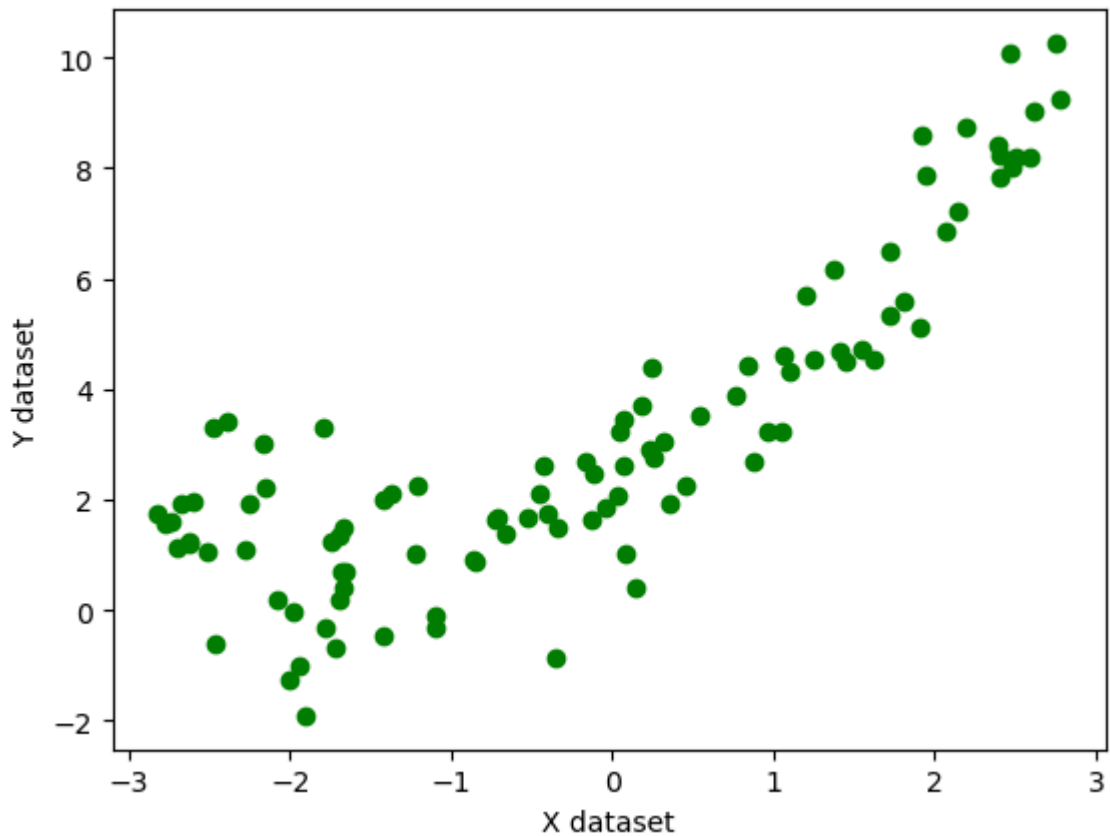
In [5]: y

```
Out[5]: array([[ 2.26272825],
               [-0.32685047],
               [ 5.60479394],
               [ 1.73386708],
               [ 9.02705104],
               [ 4.41642076],
               [ 8.74512232],
               [ 4.38143454],
               [ 4.52562673],
               [ 8.00503574],
               [ 4.68232524],
               [ 3.06490439],
               [ 4.32174236],
               [ 1.00851663],
               [ 0.70390413],
               [ 8.58560527],
               [ 3.24098166],
               [ 3.53445121],
               [ 2.91829855],
               [ 1.01457072],
               [ 3.29450271],
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               [ 7.20460166],
               [ 0.41754642],
               [ 8.21809199],
               [ 4.54387168],
               [ 0.17468308],
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               [ 1.49079949],
               [ 2.75991714],
               [-0.31257406],
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               [ 2.10023689],
               [ 3.23576245],
               [ 3.23786839],
               [ 1.60122802],
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               [ 2.69475902],
               [ 6.85415391],
               [ 8.42299912],
               [ 1.94150249],
               [ 1.05127702],
               [10.26651813],
               [ 1.24983275],
               [10.06608267],
               [ 3.87570169],
               [-1.25090519],
               [ 2.67882389],
               [ 1.2184928 ],
               [ 6.50699373],
               [ 3.02797857],
               [ 4.49495876],
               [ 0.91864639],
               [ 2.00344218],
               [ 2.26329251],
               [ 1.50349013],
```

```
[-0.47414376],  
[-0.04335893],  
[ 5.32462405],  
[ 7.88467576],  
[ 5.70577852],  
[ 0.8609122 ],  
[ 1.36431438],  
[ 1.92482904],  
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[ 1.12954777],  
[ 1.86587079],  
[ 1.63425139],  
[ 1.5734631 ],  
[ 6.15904881],  
[ 1.63179656],  
[ 2.62190829],  
[ 5.09837442],  
[-1.91386795],  
[ 2.11493683],  
[-0.10705838],  
[ 4.6187316 ],  
[ 2.21622989],  
[-0.59613579],  
[ 1.92131586],  
[ 8.21380576],  
[ 3.43369753],  
[ 2.61271081],  
[ 1.3443366 ],  
[ 3.71906476],  
[ 1.96467148],  
[ 9.24577308],  
[ 1.10988066],  
[ 4.70780794],  
[ 7.85260266],  
[ 3.30574671],  
[ 3.42106611],  
[ 1.2310642 ],  
[ 0.39893462],  
[ 8.2107995 ],  
[-0.86958494]])
```

```
In [6]: plt.scatter(X,y,color='g')  
plt.xlabel('X dataset')  
plt.ylabel('Y dataset')
```

```
Out[6]: Text(0, 0.5, 'Y dataset')
```



```
In [7]: from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_s
```

```
In [8]: from sklearn.linear_model import LinearRegression
regression_1=LinearRegression()
regression_1.fit(X_train,y_train)
```

```
Out[8]: ▼ LinearRegression ⓘ ⓘ
LinearRegression()
```

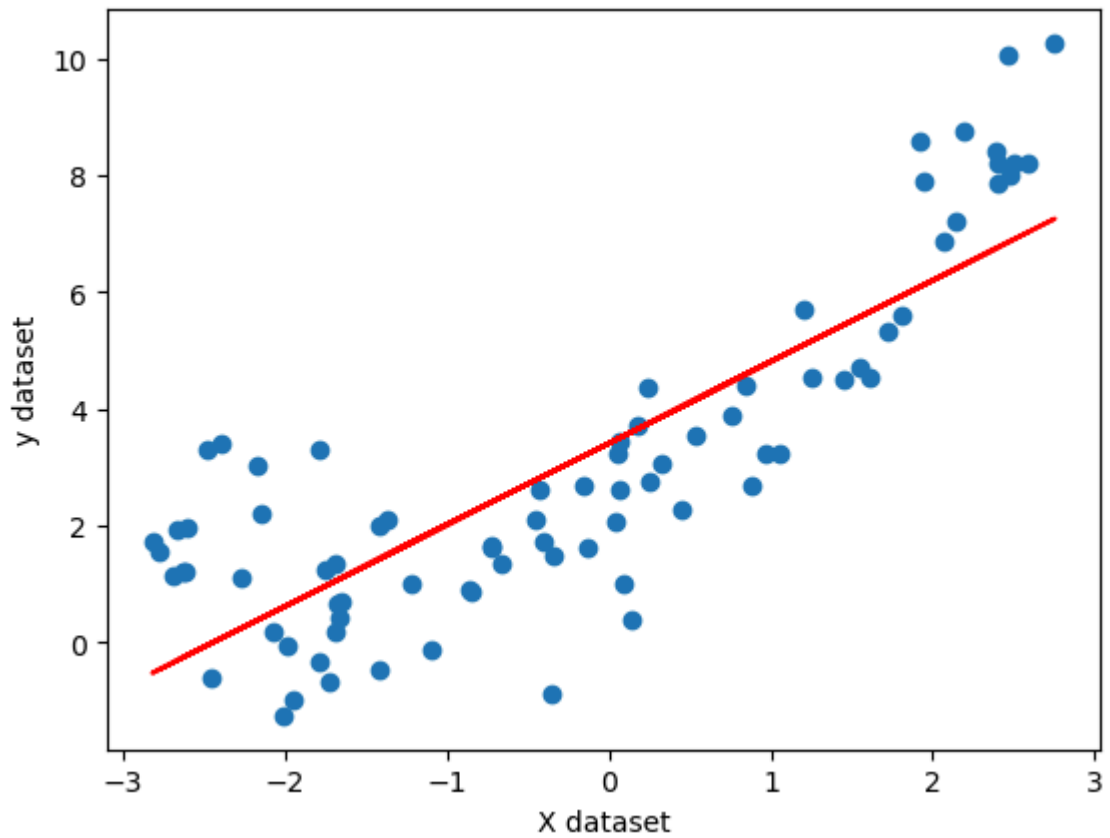
```
In [9]: from sklearn.metrics import r2_score
score=r2_score(y_test,regression_1.predict(X_test))
```

```
In [10]: print(score)
```

```
0.7453590441334617
```

```
In [11]: #visualiztion
plt.plot(X_train,regression_1.predict(X_train),color='r')
plt.scatter(X_train,y_train)
plt.xlabel("X dataset")
plt.ylabel("y dataset")
```

```
Out[11]: Text(0, 0.5, 'y dataset')
```



```
In [12]: #since linear regression fails when data is curved  
#so we use polynomial regression
```

```
In [13]: from sklearn.preprocessing import PolynomialFeatures  
poly=PolynomialFeatures(degree=2,include_bias=True)  
X_train_poly=poly.fit_transform(X_train)  
X_test_poly=poly.transform(X_test)
```

```
In [14]: X_train_poly
```

```
Out[14]: array([[ 1.00000000e+00,  1.45243762e+00,  2.10957505e+00],
 [ 1.00000000e+00,  1.81580895e-01,  3.29716214e-02],
 [ 1.00000000e+00,  1.25065832e+00,  1.56414622e+00],
 [ 1.00000000e+00,  2.06400014e+00,  4.26009657e+00],
 [ 1.00000000e+00, -2.69625501e+00,  7.26979106e+00],
 [ 1.00000000e+00,  1.92232145e+00,  3.69531975e+00],
 [ 1.00000000e+00, -1.67652677e+00,  2.81074201e+00],
 [ 1.00000000e+00, -2.62158334e+00,  6.87269923e+00],
 [ 1.00000000e+00,  2.48264388e+00,  6.16352066e+00],
 [ 1.00000000e+00, -2.78106339e+00,  7.73431360e+00],
 [ 1.00000000e+00,  3.20278813e-01,  1.02578518e-01],
 [ 1.00000000e+00, -1.75043904e+00,  3.06403684e+00],
 [ 1.00000000e+00,  6.77308076e-02,  4.58746230e-03],
 [ 1.00000000e+00, -1.69015877e+00,  2.85663666e+00],
 [ 1.00000000e+00,  2.39982527e+00,  5.75916134e+00],
 [ 1.00000000e+00,  8.43932640e-01,  7.12222301e-01],
 [ 1.00000000e+00, -6.68147516e-01,  4.46421103e-01],
 [ 1.00000000e+00, -8.48536581e-01,  7.20014330e-01],
 [ 1.00000000e+00,  3.54921242e-02,  1.25969088e-03],
 [ 1.00000000e+00,  1.05214745e+00,  1.10701425e+00],
 [ 1.00000000e+00,  7.62284066e-01,  5.81076997e-01],
 [ 1.00000000e+00, -2.82097769e+00,  7.95791514e+00],
 [ 1.00000000e+00,  2.39454116e-01,  5.73382734e-02],
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 [ 1.00000000e+00,  8.46535145e-02,  7.16621752e-03],
 [ 1.00000000e+00, -2.14972670e+00,  4.62132488e+00],
 [ 1.00000000e+00,  2.40423067e+00,  5.78032513e+00],
 [ 1.00000000e+00,  1.71822107e+00,  2.95228365e+00],
 [ 1.00000000e+00, -1.22044810e+00,  1.48949357e+00],
 [ 1.00000000e+00, -1.66720440e+00,  2.77957051e+00],
 [ 1.00000000e+00, -3.98617468e-01,  1.58895886e-01],
 [ 1.00000000e+00,  5.39622905e-01,  2.91192879e-01],
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 [ 1.00000000e+00,  1.61656944e+00,  2.61329677e+00],
 [ 1.00000000e+00, -4.49242898e-01,  2.01819182e-01],
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 [ 1.00000000e+00,  1.20334984e+00,  1.44805084e+00],
 [ 1.00000000e+00, -1.37204019e+00,  1.88249429e+00],
 [ 1.00000000e+00, -2.60351140e+00,  6.77827161e+00],
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 [ 1.00000000e+00, -2.16728004e+00,  4.69710277e+00],
 [ 1.00000000e+00,  2.38706241e+00,  5.69806694e+00],
 [ 1.00000000e+00, -2.01108896e+00,  4.04447882e+00],
 [ 1.00000000e+00, -2.67173265e+00,  7.13815536e+00],
 [ 1.00000000e+00,  2.75538120e+00,  7.59212554e+00],
 [ 1.00000000e+00, -1.72163424e+00,  2.96402444e+00],
 [ 1.00000000e+00, -1.98598454e+00,  3.94413461e+00],
 [ 1.00000000e+00,  1.41693083e-01,  2.00769299e-02],
 [ 1.00000000e+00, -1.09685704e+00,  1.20309538e+00],
 [ 1.00000000e+00,  8.81072587e-01,  7.76288904e-01],
 [ 1.00000000e+00,  4.49484773e-01,  2.02036561e-01],
 [ 1.00000000e+00,  2.46133893e+00,  6.05818933e+00],
 [ 1.00000000e+00,  2.59171718e+00,  6.71699792e+00],
 [ 1.00000000e+00, -1.41638660e+00,  2.00615100e+00],
 [ 1.00000000e+00, -4.26534441e-01,  1.81931629e-01],
 [ 1.00000000e+00,  2.50634863e-01,  6.28178346e-02],
 [ 1.00000000e+00, -2.47949416e+00,  6.14789127e+00],
 [ 1.00000000e+00, -3.46965278e-01,  1.20384904e-01],
```



```
[ 1.00000000e+00,  1.94095467e+00,  3.76730505e+00],
[ 1.00000000e+00,  2.50583967e+00,  6.27923246e+00],
[ 1.00000000e+00,  9.63047627e-01,  9.27460731e-01],
[ 1.00000000e+00, -1.94923723e+00,  3.79952576e+00],
[ 1.00000000e+00, -1.78591175e+00,  3.18948077e+00],
[ 1.00000000e+00, -2.62795553e+00,  6.90615028e+00],
[ 1.00000000e+00, -7.19636145e-01,  5.17876181e-01],
[ 1.00000000e+00,  1.80868505e+00,  3.27134162e+00],
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[ 1.00000000e+00, -1.30907409e-01,  1.71367496e-02],
[ 1.00000000e+00,  6.68596460e-02,  4.47021226e-03],
[ 1.00000000e+00, -2.46081753e+00,  6.05562293e+00],
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[ 1.00000000e+00, -7.27375698e-01,  5.29075405e-01],
[ 1.00000000e+00, -1.65524983e+00,  2.73985201e+00],
[ 1.00000000e+00,  1.55237529e+00,  2.40986903e+00],
[ 1.00000000e+00, -1.62490316e-01,  2.64031027e-02]])
```

In [15]: X_test_poly

```
Out[15]: array([[ 1.00000000e+00,  3.57089448e-01,  1.27512874e-01],
[ 1.00000000e+00,  1.72636103e+00,  2.98032240e+00],
[ 1.00000000e+00, -4.85820662e-02,  2.36021715e-03],
[ 1.00000000e+00, -2.51137703e+00,  6.30701458e+00],
[ 1.00000000e+00, -2.25577764e+00,  5.08853276e+00],
[ 1.00000000e+00, -2.74125073e+00,  7.51445557e+00],
[ 1.00000000e+00, -5.22429883e-01,  2.72932983e-01],
[ 1.00000000e+00,  1.06102604e+00,  1.12577625e+00],
[ 1.00000000e+00,  1.41373659e+00,  1.99865115e+00],
[ 1.00000000e+00, -1.20725939e+00,  1.45747523e+00],
[ 1.00000000e+00,  2.34407154e-01,  5.49467140e-02],
[ 1.00000000e+00, -1.21184081e-01,  1.46855816e-02],
[ 1.00000000e+00,  1.36806687e+00,  1.87160696e+00],
[ 1.00000000e+00, -1.09697935e+00,  1.20336370e+00],
[ 1.00000000e+00,  2.77923627e+00,  7.72415424e+00],
[ 1.00000000e+00,  2.61424862e+00,  6.83429586e+00],
[ 1.00000000e+00,  1.90655367e+00,  3.63494691e+00],
[ 1.00000000e+00, -1.90471326e+00,  3.62793259e+00],
[ 1.00000000e+00,  1.10336752e+00,  1.21741989e+00],
[ 1.00000000e+00, -1.66560648e+00,  2.77424496e+00]])
```

```
In [16]: from sklearn.metrics import r2_score
regression=LinearRegression()
regression.fit(X_train_poly,y_train)
y_pred=regression.predict(X_test_poly)
score=r2_score(y_test,y_pred)
print(score)
```

0.8806477565994808

In [17]: print(regression.coef_)

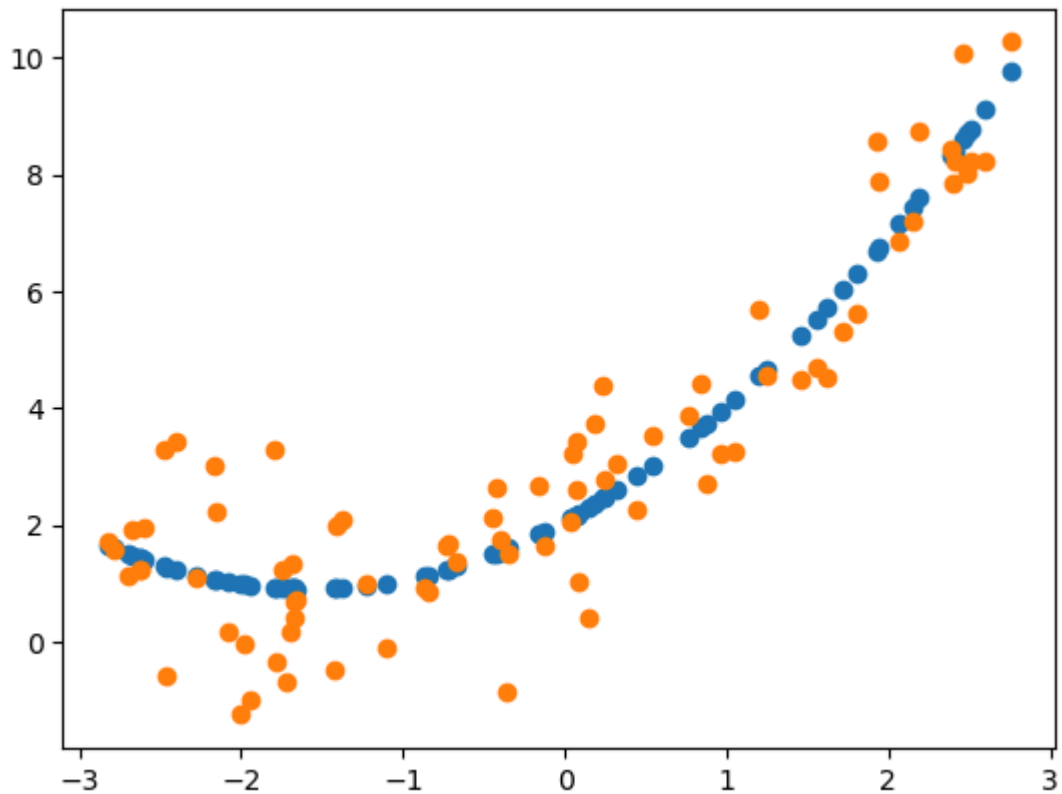
[0. 1.48895215 0.47565722]]

In [18]: print(regression.intercept_)

[2.06845024]

```
In [19]: plt.scatter(X_train, regression.predict(X_train_poly))  
plt.scatter(X_train, y_train)
```

```
Out[19]: <matplotlib.collections.PathCollection at 0x1296c78f0>
```



```
In [20]: poly=PolynomialFeatures(degree=3,include_bias=True)  
X_train_poly=poly.fit_transform(X_train)  
X_test_poly=poly.transform(X_test)
```

```
In [21]: X_train_poly
```

```
Out[21]: array([[ 1.00000000e+00,  1.45243762e+00,  2.10957505e+00,
                  3.06402616e+00],
                [ 1.00000000e+00,  1.81580895e-01,  3.29716214e-02,
                  5.98701651e-03],
                [ 1.00000000e+00,  1.25065832e+00,  1.56414622e+00,
                  1.95621248e+00],
                [ 1.00000000e+00,  2.06400014e+00,  4.26009657e+00,
                  8.79283992e+00],
                [ 1.00000000e+00, -2.69625501e+00,  7.26979106e+00,
                 -1.96012106e+01],
                [ 1.00000000e+00,  1.92232145e+00,  3.69531975e+00,
                  7.10359241e+00],
                [ 1.00000000e+00, -1.67652677e+00,  2.81074201e+00,
                 -4.71228422e+00],
                [ 1.00000000e+00, -2.62158334e+00,  6.87269923e+00,
                 -1.80173538e+01],
                [ 1.00000000e+00,  2.48264388e+00,  6.16352066e+00,
                  1.53018269e+01],
                [ 1.00000000e+00, -2.78106339e+00,  7.73431360e+00,
                 -2.15096164e+01],
                [ 1.00000000e+00,  3.20278813e-01,  1.02578518e-01,
                  3.28537260e-02],
                [ 1.00000000e+00, -1.75043904e+00,  3.06403684e+00,
                 -5.36340971e+00],
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                  3.10712526e-04],
                [ 1.00000000e+00, -1.69015877e+00,  2.85663666e+00,
                 -4.82816949e+00],
                [ 1.00000000e+00,  2.39982527e+00,  5.75916134e+00,
                  1.38209809e+01],
                [ 1.00000000e+00,  8.43932640e-01,  7.12222301e-01,
                  6.01067646e-01],
                [ 1.00000000e+00, -6.68147516e-01,  4.46421103e-01,
                 -2.98275151e-01],
                [ 1.00000000e+00, -8.48536581e-01,  7.20014330e-01,
                 -6.10958498e-01],
                [ 1.00000000e+00,  3.54921242e-02,  1.25969088e-03,
                  4.47091053e-05],
                [ 1.00000000e+00,  1.05214745e+00,  1.10701425e+00,
                  1.16474222e+00],
                [ 1.00000000e+00,  7.62284066e-01,  5.81076997e-01,
                  4.42945736e-01],
                [ 1.00000000e+00, -2.82097769e+00,  7.95791514e+00,
                 -2.24491011e+01],
                [ 1.00000000e+00,  2.39454116e-01,  5.73382734e-02,
                  1.37298856e-02],
                [ 1.00000000e+00, -2.39573445e+00,  5.73954358e+00,
                 -1.37504223e+01],
                [ 1.00000000e+00, -2.07368344e+00,  4.30016300e+00,
                 -8.91717678e+00],
                [ 1.00000000e+00,  8.46535145e-02,  7.16621752e-03,
                  6.06645499e-04],
                [ 1.00000000e+00, -2.14972670e+00,  4.62132488e+00,
                 -9.93458547e+00],
                [ 1.00000000e+00,  2.40423067e+00,  5.78032513e+00,
                  1.38972350e+01],
                [ 1.00000000e+00,  1.71822107e+00,  2.95228365e+00,
                  5.07267597e+00],
                [ 1.00000000e+00, -1.22044810e+00,  1.48949357e+00,
                 -1.81784961e+00],
```

[1.00000000e+00, -1.66720440e+00, 2.77957051e+00,
-4.63411218e+00],
[1.00000000e+00, -3.98617468e-01, 1.58895886e-01,
-6.33386755e-02],
[1.00000000e+00, 5.39622905e-01, 2.91192879e-01,
1.57134347e-01],
[1.00000000e+00, 4.75938758e-02, 2.26517702e-03,
1.07808554e-04],
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4.22457571e+00],
[1.00000000e+00, -4.49242898e-01, 2.01819182e-01,
-9.06658341e-02],
[1.00000000e+00, 2.18882298e+00, 4.79094603e+00,
1.04865328e+01],
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In [22]: X_test_poly

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```

```
In [23]: from sklearn.metrics import r2_score
         regression=LinearRegression()
         regression.fit(X_train_poly,y_train)
         y_pred=regression.predict(X_test_poly)
         score=r2_score(y_test,y_pred)
         print(score)
```

```
0.8831467171731635
```

```
In [24]: X_new=np.linspace(-3,3,200).reshape(200,1)
         X_new_poly=poly.transform(X_new)
```

```
In [25]: X_new_poly
```

```
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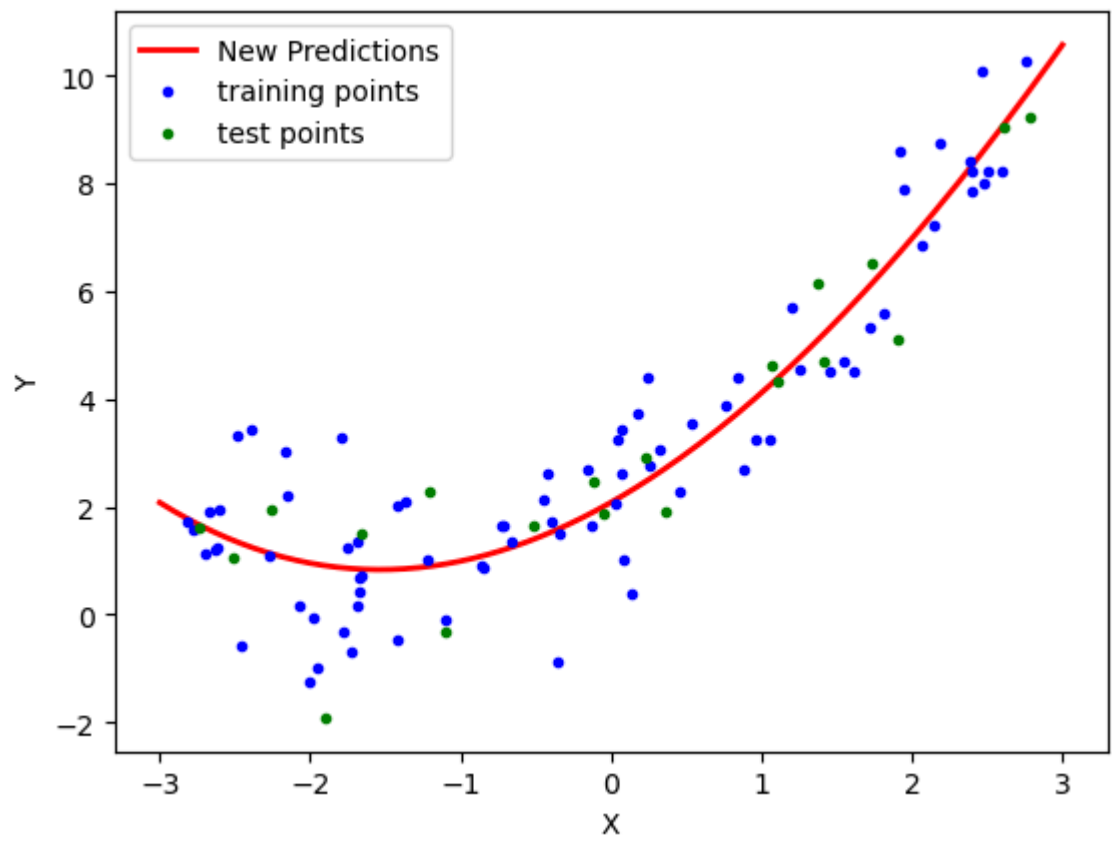
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```
In [26]: y_new=regression.predict(X_new_poly)
plt.plot(X_new,y_new,'r-',linewidth=2,label='New Predictions')
plt.plot(X_train,y_train,'b.',label='training points')
plt.plot(X_test,y_test,'g.',label='test points')
plt.xlabel('X')
plt.ylabel('Y')
plt.legend()
plt.show()
```



In []: